Other characterizations of OWPuzzs. [CGGH25] and [HM24] showed the equivalence between the existence of OWPuzzs and hardness of estimating Kolmogorov complexity. These two papers and [KT25] also showed the equivalence between the existence of OWPuzzs and hardness of estimating quantum probability.

1.3 Technical Overview

In this subsection, we provide a high-level overview of the proofs of our main results.

Proof of Theorem 1.1. Let us first explain our proof of Theorem 1.1, which is the equivalence between OWPuzzs and average-case hardness of proper quantum distribution learning. We first explain how to construct OWPuzzs from average-case hardness of proper quantum distribution learning. To show it, we first assume average-case hardness of proper quantum distribution learning. This means that there exist a QPT algorithm $\mathcal{S}$ that samples a hidden parameter z, and a QPT algorithm $\mathcal{D}$ that takes z as input, and outputs a bit string x. The hardness of learning requires that, for any polynomial t and any QPT algorithm that receives $x_1,\ldots,x_t$ independently sampled from $\mathcal{D}(z)$, it is hard to find z^* such that $\mathsf{SD}(\mathcal{D}(z^*),\mathcal{D}(z))$ is small.

We construct a OWPuzz, $(\mathsf{Samp},\mathsf{Vrfy})$, from such $\mathcal{S}$ and $\mathcal{D}$ as follows. Samp algorithm samples z from $\mathcal{S}$ and independently samples $x_1,\ldots,x_t$ from $\mathcal{D}(z)$. It then outputs $\mathsf{ans}:=z$ and $\mathsf{puzz}:=\{x_i\}_{i\in[t]}$. Next, Vrfy algorithm takes $\mathsf{ans}^*=h$ and $\mathsf{puzz}=\{x_i\}_{i\in[t]}$ as input. It first computes z^* such that

$$\Pr\Big[\{x_i\}_{i\in[t]}\leftarrow\mathcal{D}(z^*)\Big]=\max_a\Pr\Big[\{x_i\}_{i\in[t]}\leftarrow\mathcal{D}(a)\Big]. \tag{1}$$

Then, Vrfy outputs 1 if $\mathcal{D}(z^*)$ is statistically close to $\mathcal{D}(h)$, and outputs 0 if $\mathcal{D}(z^*)$ is statistically far from $\mathcal{D}(h)$.

We can show that, $\mathcal{D}(z^*)$ and $\mathcal{D}(z)$ are statistically close with high probability over $z\leftarrow\mathcal{S}$ and $x_1,\ldots,x_t\leftarrow\mathcal{D}(z)^{\otimes t}$. Therefore, in order to output h such that $1\leftarrow\mathsf{Vrfy}(\{x_i\}_{i\in[t]},h)$, the adversary must produce h such that $\mathcal{D}(h)$ is close to $\mathcal{D}(z)$. However, this is prohibited by the average-case hardness of proper quantum distribution learning. Therefore, the security of the OWPuzz follows.

Next, let us explain how to show average-case hardness of proper quantum distribution learning from OWPuzzs. [KT24, CGG24] showed that OWPuzzs imply non-uniform QPRGs (nuQPRGs). Therefore, it is sufficient to show average-case hardness of proper quantum distribution learning from nuQPRGs. A nuQPRG is a QPT algorithm Gen that takes an advice string $\mu\in[n]$ as input, and outputs n-bit strings. The security guarantees that there exists a $\mu^*\in[n]$ such that $\mathsf{Gen}(\mu^*)$ is statistically far but computationally indistinguishable from the uniform distribution U_n.

We show the average-case hardness of proper quantum distribution learning from nuQPRGs, Gen. We construct a pair $(\mathcal{S},\mathcal{D})$ of QPT algorithms as follows. $\mathcal{S}$ samples $\mu\leftarrow[n]$ and $b\leftarrow\{0,1\}$, and outputs (μ,b). $\mathcal{D}$ takes (μ,b) as input, and outputs (μ,x), where $x\leftarrow U_n$ if $b=1$, and $x\leftarrow\mathsf{Gen}(\mu)$ if $b=0$. For the sake of contradiction, suppose that $(\mathcal{S},\mathcal{D})$ is not average-case hard to learn. This means that there exists a QPT algorithm $\mathcal{A}$, given samples $(\mu,x_1),...,(\mu,x_t)$ independently sampled from $\mathcal{D}(\mu,b)$ (with $(\mu,b)\leftarrow\mathcal{S}$), that can output (μ^*,b^*) such that $\mathcal{D}(\mu^*,b^*)$ is statistically close to $\mathcal{D}(\mu,b)$.

Because the first output of $\mathcal{D}(\mu,b)$ is μ, the first output μ^* of $\mathcal{A}$ has to be equal to μ: otherwise, $\mathsf{SD}(\mathcal{D}(\mu,b),\mathcal{D}(\mu^*,b^*))$ cannot be small. Moreover, if μ is such that $\mathsf{Gen}(\mu)$ is statistically far from the uniform distribution, $\mathcal{A}$'s second output b^* has to be equal to b, because otherwise $\mathsf{SD}(\mathcal{D}(\mu,b),\mathcal{D}(\mu,b^*))$ cannot be small.

Therefore, from such $\mathcal{A}$, we can construct a QPT algorithm $\mathcal{B}$ that breaks the security of Gen as follows. Let Good be the set of all $\mu\in[n]$ such that $\mathsf{Gen}(\mu)$ is statistically far from U_n. Our construction of $\mathcal{B}$ is as follows: On input $x_1,...,x_t$ independently sampled from $\mathsf{Gen}(\mu^*)$ or U_n, $\mathcal{B}$ runs $\mathcal{A}((\mu,x_1),...,(\mu,x_t))$ for